



Erick Díaz

Data Scientist

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Technical Expertise

Machine Learning & Deep Learning
NLP & Generative AI
MLOps
Federated Learning
Bioinformatics

Education

Master's Degree in Mathematical Engineering and Computing | 2020–2022
UNIR - Universidad Internacional de La Rioja

Big Data and Artificial Intelligence Specialization | 2020–2021
UNIR - Universidad Internacional de La Rioja

Bioinformatics Specialization | 2019
Universidad San Carlos de Guatemala

Computer Science Engineer | 2013
Universidad San Carlos de Guatemala

Skills

Core Expertise: Machine Learning, NLP, LLMs, Deep Learning, Computer Vision

Languages: Python (7 yrs), PySpark, R

ML Frameworks: TensorFlow, PyTorch, Scikit-learn

Big Data: Apache Spark, Hadoop, Hive, HDFS

Databases: SQL (6 yrs), Data Warehousing

Human Languages: English, Spanish

Work Experience

- Jul 2022 Today Sr. Data Scientist — Vana | Remote 3 yrs
- Technical Lead of the DS team, designing and overseeing data science projects.
 - Create and maintain risk data science services.
 - Build models for micro-loan acceptance.
- Jan 2025 Today Lecturer & Researcher — USAC, Escuela de Ciencias Físicas y Matemáticas
- Mar 2021 Jul 2022 Sr. Data Engineer/Data Scientist — Zubale | Remote 1 yr 5 mos
- Developed a forecasting model for in-store demand (100+ stores).
 - Assessed and interviewed new members of the DS team.
 - Collaborated on a Python library for ETLs.
- Jan 2020 Mar 2021 Data Scientist — HealthCare.inc | Guatemala 1 yr
- ETL development using Python and SQL.
 - ML model for cost and revenue forecasting using PyTorch.
- Apr 2018 Dec 2019 Data Scientist 1 — Xoom, a PayPal Service | Guatemala 1 yr 9 mos
- Data wrangling and pipelines for ML models using PySpark and TensorFlow.
- Feb 2016 Mar 2018 Big Data Engineer — Xoom, a PayPal Service | Guatemala 2 yrs 2 mos
- Developed data pipelines using PySpark and Hadoop.

Research Projects and Publications

Development of a computer vision system for automatic detection and classification of *P. Vivax* malaria parasites in microscopic blood smear images. Used CNNs (U-Net, YOLO) in TensorFlow on a dataset of 1,364 annotated images (80K+ labeled cells). Presented at ICLR 2020 Workshop.

GitHub Repo · ICLR 2020 Workshop Poster

Select Certifications

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|-----------|---|-----------------|
| 2025 | Retrieval Augmented Generation (RAG) | Coursera |
| 2024 | Generative AI with Large Language Models (LLMs) | Coursera |
| 2020 | Mathematics for Machine Learning Specialization | Coursera |
| 2020 | Fundamentals of Reinforcement Learning | Coursera |
| 2019 | Intro to TensorFlow for AI, ML & Deep Learning | Coursera |
| 2018 | Machine Learning | Coursera |
| 2017 | Neural Networks and Deep Learning | deeplearning.ai |
| 2016–2017 | Big Data, Hadoop & Apache Spark | Coursera / Edx |

